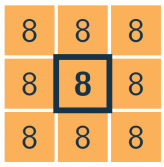
Local Moran's I: do I resemble my neighbors?

Focus on the **outlined center county**. All eight surrounding counties are neighbors.

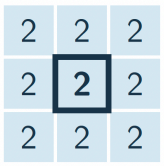
Overall study-area mean = 5 • 8 = high • 2 = low

High–High



Similar values
Positive I_i

Low–Low



Similar values
Positive I_i

High–Low



Spatial outlier pattern
Negative I_i

Low–High



Spatial outlier pattern
Negative I_i

Ask: Does positive local Moran's I always mean high values?

These are illustrative neighborhoods within a larger study area.
Permutation tests assess significance before labeling clusters or outliers.

```r
library(spdep)

values <- c(
  8, 8, 8,
  8, 2, 8,
  8, 8, 8
)

# Neighbors share an edge or a corner, as in the slide
neighbors <- cell2nb(3, 3, type = "queen")
weights <- nb2listw(neighbors, style = "W")

# Calculate local Moran's I
result <- localmoran(values, weights)

# Show the statistic for the center county (position 5)
result[5, "Ii"]
```